

ORTHOGONAL MATRIX PACKAGE PROGRAM (ORTMATPAC) FOR THE WANG-700

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**TECHNICAL REPORT
NO. 24**

PREFACE

In order to make our extensive series of technical reports more readily available, we have scanned the old master copies and produced electronic versions in Portable Document Format. The quality of the images varies depending on the quality of the originals. The images have not been converted to searchable text.

ORTHOGONAL MATRIX PACKAGE PROGRAM

(O R T M A T P A C)

FOR THE WANG-700

By E. Dorrer

ORTMATPAC is a first attempt to make available to users of the WANG 700 desk calculator a few **routines** for fundamental analytical-geometric and analytical-photogrammetric problems.

Basically three different types of routines are contained:

- (1) Trigonometric functions, such as , SIN, COS, ARCTAN, together with conversion functions from degrees, minutes, seconds or from gons to radians and vice versa. These functions were taken from SURVTRIGPACK, the idea behind them being to use only radians for angles during any type of computations. Degrees or gons are required only for I/O-operations.
- (2) Storing and recalling elements of vectors and matrices. Presently, matrices are confined to size 3×3 . It was considered as fundamentally important to keep these two routines as general as possible; i.e., to influence from outside the position of the arrays to be acted upon.
- (3) For manipulating spatial rotations at the present stage two routines for determining an orthogonal 3×3 matrix from 3 independent orientation angles, and vice versa, are available. Although similar programs could have been incorporated relatively easy, it was felt to make use of the sequence (ω, ϕ, κ) of rotations only. Whereas for aerial photogrammetric purposes the sequence as such is mostly immaterial, terrestrial photogrammetric formulation require it that way. At a later time, similar routines for the sequence (ω, ϕ, κ) and for the Rodriguez parameters are intended to be included.

It is hoped that the program package will serve some needs of the photogrammetrist who wants quick results for his problems that are not worthwhile to be solved by larger computers.

Egon Dorrer
April 1972

WANG 700A PROGRAM DESCRIPTION

Programmer: E. Dorrer

Date: March 1972

Program Title

Verify Program: 6182

ORTHOGONAL MATRIX PACKAGE PROGRAM

Set P.C. 000

(O R T M A T P A C)

No. of Steps 608

Data Registers used 000, 001, 002, 003, 004, 120, 121.

Marks used External: 0000 - 0006, 0012 - 0015.

Internal: 0201 - 0213, 0300, 0315, 1402, 1403,
1606 - 1508, 1513, 1515.

Operating Procedure

1. Load the program tape;

Depress REVIND

Depress TAPE READY, RUN, PRIME, LOAD PROGRAM.

2. PRIME; VERIFY PROGRAM; Read $x = 6182$.

3. PRIME.

Then load data according to instruction on the following pages.

Function Keys

0000	RADEGR	0001	RADGON
0002	SIN	0003	COS
0004	MATSTIN	0005	ORTMAT2
0006	ROTANG2		
0012	MATREIN	0013	ARCTAN
0014	GON	0015	DEGR

Function Key: 00 Purpose: To convert angles given
 RADEGR in degrees, minutes, and
 seconds to radians.

Function: $X = \text{RADEGR } (X)$

Procedure: Key DMS-value in X; Depress 00; Read Radians in X.

Example: Given $\alpha = -35^\circ 17' 03''$
 Key -351703.6 in X; Depress 00
 Read -.615827791037

Function Key: 01 Purpose: To convert angles given in
 RADGON Gon to Radians.

Function: $X = \text{RADGON } (X)$

Procedure: Key Gons in X; Depress 01.
 Read Radians in X.

Example: Given $\alpha = 103.6782$ gon
 Key 103.6782 in X; Depress 01
 Read +1.62857335728

Function Key: 02 Purpose: To compute the sine of an
 SIN angle given in radians.

Function: $X = \text{SIN } (X)$

Procedure: Key radians in X; Depress 02.

Read SIN of X in X.

Example: Key 0.571 in X; Depress 02

Read +.540518039256

Function Key: 03 Purpose: To compute the cosine of
 COS an angle given in radians.

Function: $X = \text{COS } (X)$

Procedure: Key radians in X; Depress 03.

Read cos of X in X.

Example: Key 0.571 in X; Depress 03.

Read +.841360922257

Function Key: 04 Purpose: To store indirectly in core
 MATSTIN elements of a vector or of

Function: $((000)+Y) = \text{MATSTIN } (000, X, Y)$ a 3 x 3 matrix (stored row-wise).

Procedure: (1) Store starting address of array in register 000.

PRIME; Key starting address in X;

 Depress STIN.

Note: (2) For vector element V_i

Starting Key i in Y; Key V_i in X
address must

be larger For matrix-element M_{ij}

than 004. Key $i \cdot j$ in Y; Key M_{ij} in X

 Depress MATSTIN

 (3) For other elements go to (2).

Example: The element $M_{2,3} = 216.1$ of a 3 x 3 matrix is supposed to

be stored in an array starting with address 015. Then,

PRIME; Key 015 in X; Depress STIN

Key 2.3 in X; Depress ↑; Key 216.1 in X.

Depress MATSTIN.

216.1 is now stored in address 020.

Function Key: 05 Purpose: To compute the elements of
 ORTMAT2 a 3 x 3 orthogonal matrix [r]
 from three rotation angles
 $\alpha_1, \alpha_2, \alpha_3$, where α_2 is primary
 (ϕ) and α_1 is secondary (ω)
 rotation. The result is
 stored according to MATSTIN
 directly behind the stored
 3 angles.

Procedure: (1) Store $\alpha_1, \alpha_2, \alpha_3$ in consecutive addresses.
 (2) Key starting address of α -array in X;
 Depress ORTMAT2.
 (3) Result can be displayed immediately with MATREIN
 (see function Key 12), since starting address of [r]
 is supplied in 000.

Note: The rotation angles must be given in radians.

Storage: α_1 α_2 α_3 r_{11} r_{12} r_{13} r_{21} r_{22} r_{23} r_{31} r_{32} r_{33}
 $\uparrow$ $\uparrow$
) α () r (=) α (+ 3

Example: (1) Assume (008) = 0, (009) = .5, (010) = -.7
 (2) Key 008 in X; Depress 05;
 (Read 3 in X; 19 in Y; both not significant)
 (3) Result is stored in (011) - (019);
 .671212166160 .565354208381 .479425538599
 -.644217687233 .764842187280 0
 -.3666848775582 -.308854411678 .877582561891

Function Key: 06

ROTANG2

Purpose: To compute the three

independent rotation parameters

$\alpha_1, \alpha_2, \alpha_3$ of a 3 x 3 orthogonal

matrix [r], stored row-wise in

a core array. The result is

stored immediately prior to [r].

Primary rotation is α_2 (φ),

secondary is α_1 (ω).

Procedure: (1) Store [r] row-wise in consecutive addresses.

(2) Key starting address of [r] - 3 in X

Depress ROTANG2.

(3) Result in radians can be displayed immediately
with MATREIN (see function Key 12).

Example: (1) Assume the result of the example of
ORTMAT2 is input here.

(2) Key 008 in X; Depress 06;

(Read 0 in X; 8 in Y)

(3) Result stored in (008) - (010):

0, .499999999989, -.699999999997.

Function Key: 12
MATREIN

Purpose: To indirectly recall the
elements of a vector or of
a 3 x 3 matrix, stored
row-wise.

Function: (X,Y) = MATREIN (000, X)

Operating Procedure: (1) Store starting address of array in address 000,
e.g. by:

PRIME; Key starting address in X; Depress STIN.

(2) For vector-element V_i Key i in X;

For matrix-element M_{ij} Key i·j in X;

Depress MATREIN

Read V_i (or M_{ij}) in X; i (or i·j) in Y.

(3) For other element goto (2).

Example: Assume that matrix [M] to be recalled is stored in locations 005-013.

PRIME; Key 005 in X; Depress STIN.

In order to recall element $M_{1,2}$,

Key 1.2 in X, Depress MATREIN.

Read $M_{1,2}$ in X and 1.2 in Y.

Storage: $\alpha_1 \quad \alpha_2 \quad \alpha_3 \quad r_{11} \quad r_{12} \quad r_{13} \quad r_{21} \quad r_{22} \quad r_{23} \quad r_{31} \quad r_{32} \quad r_{33}$
 $\uparrow \quad \quad \quad \uparrow$
 $)\alpha(\quad \quad \quad)r(=)\alpha(+ 3$

ARCTAN

Purpose: To compute the arctan
of a value.

Function: $X = \text{ARCTAN}(X)$

Procedure: Key argument in X; Depress 13.

Read arctan given in radians in X.

Example: Key 1.0073821 in X; Depress 13.

Read .789075623072

Note: If result is wanted in degrees, minutes, seconds, depress DEGR.

Function Key: 14

GON

Purpose: To convert an angle given in radians into gons.

Function: $X = \text{GON}(X)$

Procedure: Key Radians in X; Depress 14.

Example: Key 0.3781259; Depress 14. Read 24.0722424384

Function Key: 15

DEGR

Purpose: To convert an angle given in radians into a DMS-value (degrees, minutes, seconds)

Function: $X = \text{DEGR}(X)$

Procedure: Key radians in X; Depress 15

Read DMS-value in X.

Example: Key 0.3781259; Depress 15. Read 213954.065500.
(2, 39' 54", 065500)

ORTMATPAC

Storage Map

Routines

Register	0000	0001	0002	0003	0004	0005	0006	0012	0013	0014	0015
121	X	X	X	X		(X)	(X)		X	X	X
120	X		X	X	(X)	(X)	(X)	(X)	X		X
000					X	X	X	X			
001						X	X	X			
002						X	(X)		X		
003			X	X		(X)	(X)		X		
004			X	X		(X)	(X)		X		

Note: Subroutine 0300 = ADDR is a useful subprogram for computing indirectly addresses of array elements

000	MARK		
001	RADEGR	0000	BEGIN RADEGR
002	↑		
003	WRITEA		
004	STDx		
005	INTx		
006	STDx		
007	120		
008	6		
009	0		
010	*DX		
011	120		
012	+		
013	WRITEA		
014	*DX		
015	INTx		
016	+DX		
017	120		
018	4		
019	0		
020	*DX		
021	120		
022	REDx		
023	120		
024	-		
025	PI		
026	*		
027	6		
028	4		
029	8		
030	SEXP		
031	6		
032	/		
033	+		
034	SEARCH		END RADEGR
035	1515		
036	MARK		
037	RADGON	0001	BEGIN RADGON
038	↑		
039	PI		
040	*		
041	2		
042	0		
043	0		
044	/		
045	+		
046	SEARCH		END RADGON
047	1515		

048	MARK		
049	COS	0003	BEGIN COS
050	+		
051	+		
052	PI		
053	+		
054	2		
055	/		
056	-		
057	MARK		
058	SIN	0002	BEGIN SIN
059	STDx		
060	120		
061	PI		
062	+		
063	+		
064	REDx		
065	120		
066	EXY		
067	/		
068	EXY		
069	INTx		
070	*		
071	REDx		
072	120		
073	EXY		
074	-		
075	PI		
076	/		
077	-		
078	INTx		
079	WRITEA		
080	LOGEX		
081	-		
082	CHSGN		
083	1		
084	*		
085	.		
086	5		
087	SKIP Y<X		
088	SEARCH		
089	1508		
090	CHSGN		
091	SKIP Y>=X		
092	SEARCH		
093	1508		
094	MARK		
095	1507		

096	PI	
097	*	
098	STDY	0414
099	120	
100	-	
101	X+2	
102	STDY	
103	003	
104	1	
105	5	
106	+	
107	1	
108	STDY	
109	004	
110	MARK	
111	1506	
112	REDX	
113	003	
114	CHSGN	
115	*DX	
116	004	
117	-	
118	/DX	
119	004	
120	1	
121	-	
122	-	
123	/DX	
124	004	
125	1	
126	+DX	
127	004	
128	RESID	
129	+DX	
130	004	
131	1	
132	-	
133	SKIP Y=X	
134	SEARCH	
135	1506	
136	REDX	
137	004	
138	REDY	0415
139	120	
140	*	
141	-	
142	SEARCH	
143	1515	
144	MARK	

145	1508	
146	-	
147	-	
148	-	
149	CHSGN	
150	↑	
151	SEARCH	
152	1507	END COS AND SIN
153	MARK	
154	ARCTAN 0013	BEGIN ARCTAN
155	WRITEA	
156	CLX	
157	↑	
158	.	
159	5	
160	SKIP Y>=X	
161	WRITEA	
162	X+2	
163	1	
164	+	
165	STDY 0414	
166	003	
167	2	
168	-	
169	REDX	
170	003	
171	/	
172	-	
173	STDY 0414	
174	002	
175	*	
176	STDY 0414	
177	003	
178	1	
179	STDX	
180	120	
181	1	
182	5	
183	↑	
184	8	
185	STDX	
186	004	
187	MARK	
188I	1513	
189	REDX	
190	003	
191	*DX	
192	120	

193	REDX		
194	004		
195	*DX		
196	004		
197	EXDX		
198	004		
199	*DX		
200	120		
201	←		
202	+DX		
203	120		
204	2		
205	-		
206	1		
207	-DX		
208	004		
209	EXDX		
210	120		
211	/DX		
212	120		
213	REDX		
214	004		
215	WRITEA		
216	LOGEX		
217	SEARCH		
218	1513		
219	REDY	0415	
220	002		
221	REDX		
222	120		
223	*		
224	WRITEA		
225	GO		
226	*		
227	4		
228	5		
229	WRITEA		
230	10+X		
231	WRITEA		
232	STOP		
233	*		
234	←		
235	WRITEA		
236	ENDPRGM		
237	SEARCH		
238	1515		END ARCTAN
239	MARK		
240	GON	0014	BEGIN GON

241	↑		
242	2		
243	0		
244	0		
245	*		
246	PI		
247	/		
248	←		
249	SEARCH		
250	1515		END GON
251	MARK		
252	DEGR	0015	BEGIN DEGR
253	↑		
254	WRITEA		
255	GO		
256	*		
257	CLX		
258	STDY		
259	120		
260	SUB	0315	
261	SUB	0315	
262	REDX		
263	120		
264	+		
265	←		
266	SEARCH		
267	1515		END DEGR
268	MARK		
269	SUB	0315	
270	←		
271	INTX		
272	-		
273	+DX		
274	120		
275	2		
276	10+X		
277	*DX		
278	120		
279	6		
280	0		
281	*		
282	RETURN		
283	MARK		
284	1515		
285	REDY	0415	
286	121		
287	RETURN		END TRIGPACK
288	MARK		
289	MATSTIN	0004	BEGIN MATSTIN

290	EXDX		
291	000		
292	ADDR	0300	
293	EXDX		
294	000		
295	STIN		
296	STOP		END MATSTIN
297	MARK		
298	MATREIN	0012	BEGIN MATREIN
299	↑		
300	STDY	0414	
301	001		
302	REDX		
303	000		
304	ADDR	0300	
305	REIN		
306	REDY	0415	
307	001		
308	STOP		END MATREIN
309	MARK		
310	ADDR	0300	BEGIN ADDR
311	STDY		
312	120		
313	←		
314	INTX		
315	SKIP Y=X		
316	SEARCH		
317	MATRIX	1402	
318	1		
319	SEARCH		
320	END	1403	
321	MARK		
322	MATRIX		
323	+		
324	+		
325	←		
326	INTX		
327	-		
328	EXY		
329	WRITEA		
330	1		
331	+		
332	4		
333	MARK		
334	END		
335	-		
336	REDX		
337	120		
338	+		
339	RETURN		END ADDR

340	MARK		
341	ORTMAT2	0005	BEGIN ORTMAT2
342	STDX		
343	000		
344	1		
345	SR1	0201	
346	SIN	0002	
347	CHSGN		
348	SR3	0203	
349	2		
350	.		
351	3		
352	SR2	0202	
353	3		
354	SR1	0201	
355	SIN	0002	
356	SR3	0203	
357	2		
358	.		
359	1		
360	SR2	0202	
361	3		
362	SR1	0201	
363	COS	0003	
364	SR3	0203	
365	2		
366	.		
367	2		
368	SR2	0202	
369	2		
370	SR1	0201	
371	SIN	0002	
372	SR3	0203	
373	1		
374	.		
375	3		
376	SR2	0202	
377	2		
378	SR1	0201	
379	COS	0003	
380	SR3	0203	
381	3		
382	.		
383	3		
384	SR2	0202	
385	4		
386	-		
387	SR4	0204	
388	1		
389	SR5	0205	

390	2	
391	SR6	0206
392	1	
393	SR7	0207
394	2	
395	-	
396	SR8	0208
397	8	
398	+	
399	REIN	
400	CHSGN	
401	SR3	0203
402	5	
403	-	
404	SR4	0204
405	2	
406	SR5	0205
407	1	
408	SR6	0206
409	2	
410	SR7	0207
411	1	
412	-	
413	SR8	0208
414	1	
415	+	
416	REIN	
417	CHSGN	
418	SR3	0203
419	2	
420	+	
421	SR4	0204
422	4	
423	SR5	0205
424	3	
425	SR6	0206
426	2	
427	SR7	0207
428	3	
429	+	
430	SR8	0208
431	4	
432	-	
433	REIN	
434	SR3	0203
435	1	
436	+	
437	SR4	0204
438	5	
439	SR5	0205

440	3	
441	SR6	0206
442	1	
443	SR7	0207
444	3	
445	+	
446	SR8	0208
447	1	
448	SR1	0201
449	COS	0003
450	SR3	0203
451	1	
452	.	
453	2	
454	SR10	0210
455	ADDR	0300
456	SR9	0209
457	SR9	0209
458	SR9	0209
459	3	
460	+	
461	SR9	0209
462	3	
463	+DX	
464	000	
465	STOP	END ORTMAT2
466	MARK	
467	SR1	0201
468	↑	
469	REDX	
470	000	
471	ADDR	0300
472	REIN	
473	RETURN	
474	MARK	
475	SR2	0202
476	SR10	0210
477	ADDR	0300
478	REDX	
479	001	
480	STIN	
481	RETURN	
482	MARK	
483	SR3	0203
484	STDx	
485	001	
486	RETURN	
487	MARK	
488	SR4	0204
489	REFIN	

490	*DX	
491	001	
492	RETURN	
493	MARK	
494	SR5	0205
495	+	
496	REIN	
497	STDx	
498	002	
499	RETURN	
500	MARK	
501	SR6	0206
502	-	
503	REIN	
504	*DX	
505	002	
506	RETURN	
507	MARK	
508	SR7	0207
509	SR6	0206
510	REDX	
511	002	
512	-DX	
513	001	
514	RETURN	
515	MARK	
516	SR8	0208
517	REDX	
518	001	
519	STIN	
520	RETURN	
521	MARK	
522	SR9	0209
523	1	
524	+	
525	REIN	
526	EXDX	
527	001	
528	*DX	
529	001	
530	EXDX	
531	001	
532	STIN	
533	RETURN	
534	MARK	
535	SR10	0210
536	-	
537	3	
538	EXDX	
539	000	
540	+DX	

541	000		
542	EXDX		
543	000		
544	RETURN		
545	MARK		
546	ROTANG2	0006	BEGIN ROTANG2
547	STDX		
548	000		
549	1		
550	.		
551	3		
552	SR11	0211	
553	6		
554	SR12	0212	
555	2		
556	SR13	0213	
557	2		
558	.		
559	1		
560	SR11	0211	
561	1		
562	SR12	0212	
563	3		
564	SR13	0213	
565	2		
566	.		
567	3		
568	SR11	0211	
569	X+2		
570	↑		
571	1		
572	-		
573	←		
574	CHSGN		
575	SQRX		
576	CHSGN		
577	/DX		
578	001		
579	1		
580	SR13	0213	
581	STOP		END ROTANG2
582	MARK		
583	SR11	0211	
584	SR10	0210	
585	ADDR	0300	
586	REIN		
587	SR3	0203	
588	RETURN		
589	MARK		
590	SR12	0212	

591	+	
592	REIN	
593	/DX	
594	001	
595	RETURN	
596	MARK	
597	SR13	0213
598	↑	
599	REDX	
600	000	
601	ADDR	0300
602	REDX	
603	001	
604	ARCTAN	0013
605	STIN	
606	RETURN	
607	ENDPRG	END ORTMATPACK.

SURVEY TRIGONOMETRIC PACKAGE PROGRAM

(S U R V T R I G P A K)

For the WANG-700

By E. Dorrer

Introduction 1

SURVTRIGPAK is an attempt to utilize the WANG-700 programmable desk calculator more efficiently for simple, plane-surveying problems such as conversion of angles from degrees, minutes, seconds in radians and vice versa, or from gons (=grads) in radians and vice versa, the essential trigonometric functions sin, cos, tan, arctan, etc. In addition frequently occurring problems such as conversion from polar to rectangular (in the plane), or from spherical to cartesian (in space) and vice versa, have been included for further convenience.

It has been the intention to use as arguments for the trigonometric functions angles given in radians only. This makes the program package general, quite flexible and independent of special conversion conventions. Angles given in degrees, minutes and seconds are supposed to be keyed in as one number (DMS). This arrangement is much more convenient than keying each of the three numbers independently. The routines ARCSIN, ARCCOS are found to be not necessary, as they can simply be expressed in terms of ARCTAN, e.g.

$$\text{ARCSIN } X = \text{ARCTAN } \frac{X}{\sqrt{1 - X^2}}$$

$$\text{ARCCOS } X = \pi/2 - \text{ARCSIN } X$$

SURVTRIGPAK has been hastily and quickly compiled. The form of this report may therefore be somewhat undesirable, although the program package has been thoroughly tested. Also, the documentation is principally in order. In a University environment, form per se is often not as

important as content and exactness are.

Particular thanks has to be given to Mr. O. Afolabi, who tested the program package and made it working properly. Without him, SURVTRIGPAK would have remained unfinished fragments (dated back to 1971).

It is hoped that this program package will serve some of the needs of practicing surveyors, particularly when the work involved is too small as to require a large electronic data processor.

Egon Dorrer
March 1973

Introduction 2

A Surveying Engineer is often involved in carrying out calculations in one form or the other. If the calculation is to be performed only once, it is simple enough to do it manually on the key board of any one of the calculating machines at his disposal. However, if the same calculations are to be done repeatedly, it is highly beneficial to record and save the steps of the calculation in the form of a program and let the calculator perform these repeated operations. The program is loaded into the core of the calculating machine and executed from the core. In WANG 700, such programs can be stored on magnetic tape for later use.

The routines contained in SURVEY-TRIGONOMETRIC PACKAGE (SURVTRIGPAK) are of four main types:

- (1) Trigonometric functions, such as, SIN, COS, TAN, ARCTAN, together with conversion functions from degrees, minutes and seconds or from gon to radians (RAD) and vice versa. Degrees or grades are required only for I/O-operations.
- (2) Routines to convert polar coordinates to rectangular coordinates and vice versa. This is extremely useful in calculations of traverse coordinates and conversely to obtain azimuths and distances from known coordinates.
- (3) Routine ECCORR may be used to reduce bearings observed at eccentric stations to centre. The routine takes care of the two possible cases; that is: (a) when the distances from the eccentric station to targets are known and (b) when distances from centre to targets are known.

- (4) Routine INSECT calculates the coordinates of the point of intersection of two observed azimuths from two given points. Routine TRAVERS computes coordinates of traverse stations when traverse angles and distances are keyed in sequentially.

WANG 70CA PROGRAM DESCRIPTION

Programmer: E. DORRER

Date:

Program Title

Verify Program: 7156

SURVEY TRIGONOMETRIC PACKAGE PROGRAM Set P.C. 000

(S U R V T R I G P A C)

No. of Steps 629

Data Registers used 000, 001, 002, 003, 004, 005.

Maps used External:

Internal:

(a) TO 'LEARN' PROGRAM (keyboard to core)

1. Set LEARN mode
2. SET PC to desired step
3. Index leaps in program
4. Index END PROGRAM as last step in program.

(b) TO RECORD PROGRAM (core to tape)

1. Insert tape cartridge, REWIND if necessary
2. Set TAPE READY
3. SET PC to first step of program
4. Index RECORD PROGRAM

(c) TO LOAD PROGRAM (tape to core)

1. Set RUN mode
2. Insert tape cartridge, REWIND if necessary
3. Set TAPE READY
4. SET PC to desired step number
5. Index LOAD PROGRAM

Operating Procedure

1. Load the program tape;

Depress REWIND

Depress TAPE READY, RUN, PRIME, LOAD PROGRAM

2. PRIME; VERIFY PROGRAM; Read X = 7156

3. PRIME

Then load data according to the instruction on the following pages.

Function Keys

000	RADEGR	001	RADGON
002	SIN	003	COS
004	TAN	005	RECT
006	CART	007	ECCORR
008	INSECT	009	TRAVERS
011	SPHER	012	POLAR
013	ARCTAN	014	GON
015	DLG		

Function Key: 0000

RADEGR

Purpose: To convert angles
given in degrees,
minutes and seconds
to radians.

Function: $X = \text{RADEGR } (X)$

Procedure: Key DMS-value in X; Depress 00; Read Radians in X

Example: Given $\alpha = 65^\circ 34' 29.16$

Key 653429.16 in X; Depress 00

Read 1.14449558456

Function Key: 0001

RADGON

Purpose: To convert angles given
in GON to Radians.

Function: $X = \text{RADGON } (X)$

Procedure: Key Gons in X; Depress 01

Read Radians in X

Example: Given $\alpha = 103.6782$ gon

Key 103.6782 in X; Depress 01

Read 1.62857335728

Function Key: 0002

SIN

Purpose: To compute the sine of
an angle given in
radians.

Function: $X = \text{SIN } (X)$

Procedure: Key radians in X; Depress 02

Read SIN of X in X

Example: Key 0.571 in X; Depress 02

Read + .540473679750

Function Key: 00C3

COS

Purpose: To compute the cosine
of an angle given in
radians.

Function: $X = \text{COS}(X)$

Procedure: Key radians in X; Depress 03

Read COS of X in X

Example: Key 0.571 in X; Depress 03

Read + .841360922257

Function Key: 00C4

TAN

Purpose: To compute the tangent
of an angle given in
radians.

Function: $X = \text{TAN}(X)$

Procedure: Key radians in X; Depress 04

Read COS of X in X

Example: Key .462 in X; Depress 04

Read + .497942182120

Function Key: 0005 Purpose: To calculate rectangular
 RECT coordinates X,Y from
 polar coordinates Q,R;
 Q in radians.

Function: (X,Y) = RECT (Q,R)

Procedure: 1. Key R in X; Depress , R is in Y register
 2. Key Q in X, eventually convert into radians.
 3. Depress 0005
 4. Read Y = R sin(Q) in Y register
 Read X = R cos(Q) in X register.

Example: 1. Key 400.00 into Y register
 2. Key 300000.0 (=30°0'0") into X register
 Depress 0005
 4. Read 199.999999999 in Y
 Read 346.410161515 in X

Function Key: 0006

Purpose: To calculate three-dimensional cartesian coordinates, X, Y, Z from spherical coordinates α, β, R . α, β in radians

CART

(inverse to SPHER)

Function: 1. $(Z, R \cos \beta) = \text{CART}(\beta, R)$

2. $(X, Y) = \text{GO}(\alpha, R \cos \beta)$

Procedure:

1. Key R in Y and β in X register. Eventually convert β to radians.
2. Depress 06
3. Read Z from X-register
4. Key α in X, eventually convert to radians.
5. Depress 60
6. Read X in X and Y in Y register.

Example:

1. Key 1000.000 in Y, 152934 in X ($15^\circ 29' 34''$).
Depress 00 to convert to radians.
2. Depress 06
3. Read Z = 267.116834 from X register
4. Key 1271204 in X ($127^\circ 12' 04''$)
Depress 00 to convert to radians
5. Depress 60
6. Read X = - 582.645366199
Y = 767.576012734

Remark: R = Slope distance

β = Vertical angle

α = Horizontal angle

Function Key: 0007

ECCORR

Purpose: To convert eccentric bearings to centre given (a) distance from eccentric station to target or (b) distance from centre to target.

Function: Reference to Figure 2

$$(a) P_{ci} = f(e, P_o, S_{ci}, P_i)$$

$$(b) P_{ci} = f(e, P_o, S_{ci}, P_i)$$

Procedure:

1. Let e in Y; P_o in X
2. Depress CC
3. Depress C7
4. Let S_{ci} in Y, P_i in X

5. Depress CC

6. SEARCH 2; Read P_{ci} in X (in radians)

7. Depress C15; Read P_{ci} in X (in DMS)

Repeat steps 4-7 for different values of S_{ci} and P_i

N.B. When distance from eccentric station to target is known, procedure 4 and 6 will be

4. Let S_{ci} in Y, P_i in X

6. Let SEARCH 1, Read P_{ci} in X (in radians)

Example: Reference to Figure 2

1. Key 6.54 in Y; 000000 in X
(0° 00' 00")
2. Depress 00
3. Depress 07
4. Key 7031.0 in Y, 391247.28 in X
(39° 12' 47.28")
5. Depress 00
6. SEARCH 2; Read .684986344360 in X (in radians)
7. Depress 015; Read 391448.575601 in X (in DMS)
(39° 14' 48.575601")

For new values of S_{ci} and P_i , repeat steps 4-7

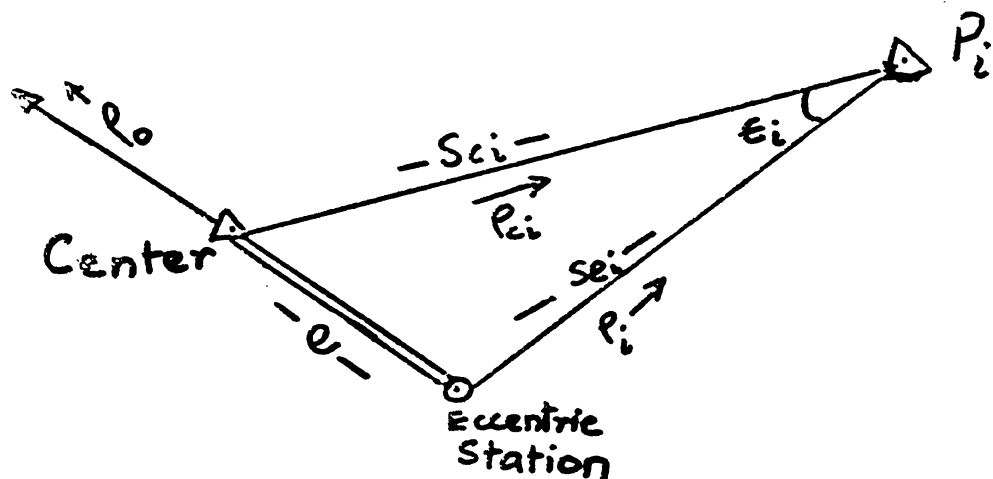
Key 5556.0 in Y, 1692747.18 in X
(169° 27' 47.18")

Depress 00

SEARCH 2, Read 2.95790440832 in X (in radians)

Depress 015, Read 1692831.57967 in X (in DMS)
(169° 28' 31.57967")

FIGURE 2



Given: (a) Fixed e ; p_0
 (b) Variable p_i ; Se_i or Sc_i

Wanted: ϵ_i or $p_{ci} = p_i + \epsilon_i$

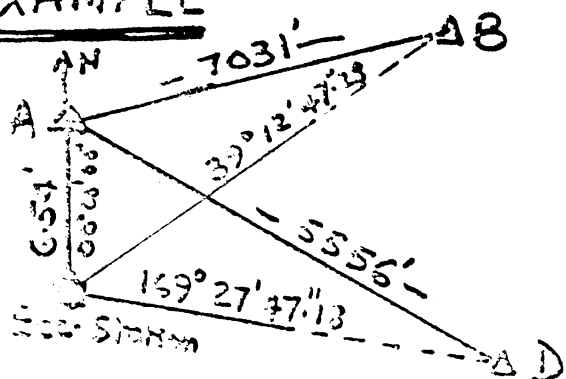
2 cases: Case 2: given Sc_i ; $p_{ci} = f(e, p_0, p_i, Sc_i)$

Case 1: given Se_i ; $p_{ci} = f(e, p_0, p_i, Se_i)$

Case 2: $p_{ci} = p_i + \arctan \left[\frac{e \sin(p_i - p_0)}{\sqrt{(Sc_i - e \sin(p_i - p_0))(Sc_i + e \sin(p_i - p_0))}} \right]$

Case 1: $p_{ci} = p_i + \arctan \left[\frac{e \sin(p_i - p_0)}{Se_i - e \cos(p_i - p_0)} \right]$

EXAMPLE



Function key: 0008

INSTRUCT

Purpose: To compute the intersection to two straight lines. That is given coordinates of A (X_1, Y_1), B (X_2, Y_2) and azimuths AC (α_{AC}) and BC (α_{BC}); required to compute C (X_C, Y_C).

Function: $(X, Y) = f(X_1, Y_1, \alpha_1, X_2, Y_2, \alpha_2)$
(See Figure 3)

Procedure:

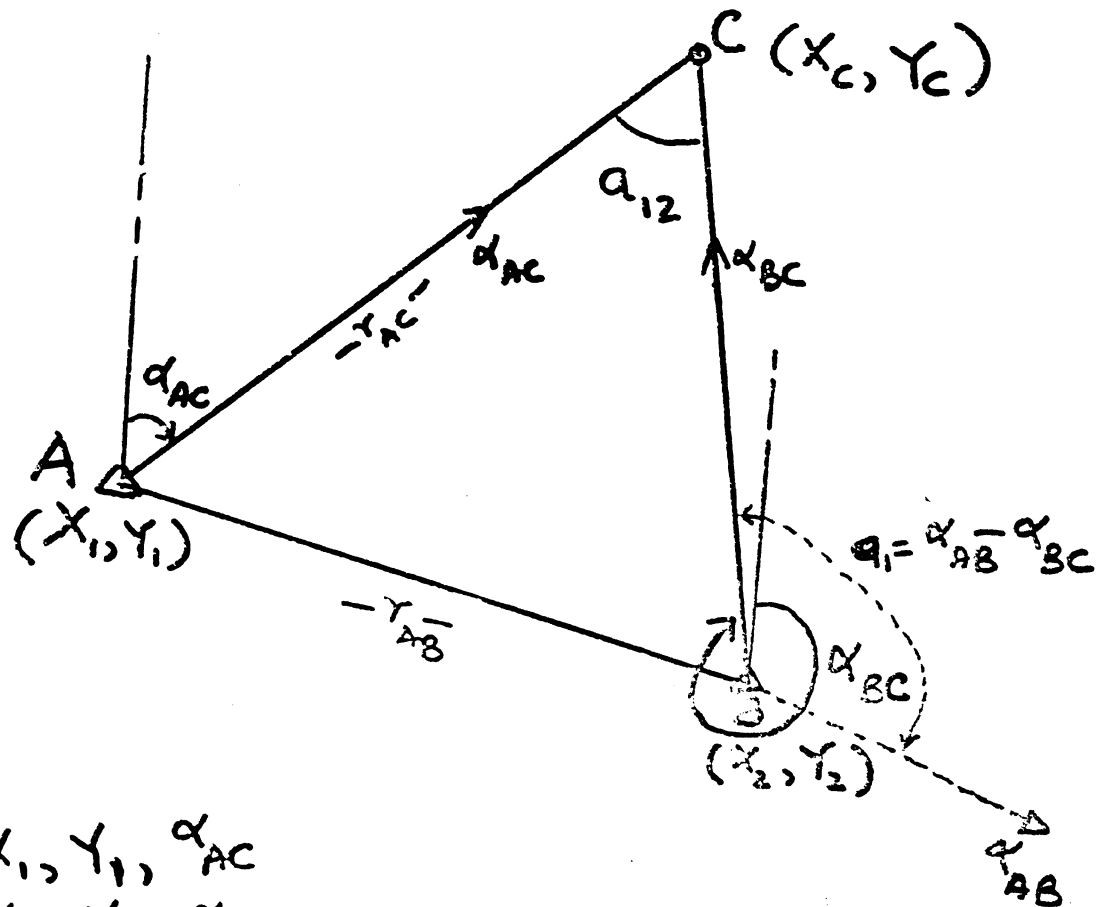
1. Depress 08
2. Read 1.0000000000 in Y; 1.0000000000 in X
3. Key Y_1 in Y; X_1 in X
4. Depress G0
5. Read 2.0000000000 in Y; 2.0000000000 in X
6. Key Y_2 in Y; X_2 in X
7. Depress G0
8. Read 1.0000000000 in Y; 2.0000000000 in X
9. Key α_1^* in Y; α_2^* in X
10. Depress G0
11. Read Y value of intersected point in Y and X value in X

* α_1, α_2 should each be converted to radians; thus before transferring α_1 to Y register, depress 00 and depress 00 after α_2 has been entered into X register. If more than one point is intersected at both stations (X_1, Y_1), (X_2, Y_2) repeat steps 9-11 for other values of α_{1i} and α_{2i} .

Example: See Figure #3

1. Depress 08
2. Read 1.0000000000 in Y; 1.0000000000 in X
3. Key 4.0 in Y; 6.0 in X
4. Depress G0
5. Read 2.0000000000 in Y; 2.0000000000 in X
6. Key 13.0 in Y, 4.0 in X
7. Depress G0
8. Read 1.0000000000 in Y; 2.0000000000 in X
9. Key 600000, depress 00, depress ↑, read
(60°00'00")
1.04719755119 in Y
Key 3300000, depress 00, read 5.75958653157
(330°00'00')
in X
10. Depress G0
11. Read 9.88397459600 in Y, 9.39711451700 in X

FIGURE 3



Given:- X_1, Y_1, α_{AC}
 X_2, Y_2, α_{BC}

Required:- X_c, Y_c

$$(\alpha_{AB}, r_{AB}) = \text{To Polar}(X_2 - X_1, Y_2 - Y_1)$$

$$\alpha_1 = \alpha_{AB} - \alpha_{BC}$$

$$\alpha_{12} = \alpha_{AC} - \alpha_{BC} = \alpha_{AC} - \alpha_{AB} + \alpha_1$$

$$r_{AC} = r_{AB} \cdot \frac{\sin \alpha_1}{\sin \alpha_{12}}$$

$$(X_c, Y_c) = (X_1, Y_1) + \text{To RECT}(\alpha_{AC}, r_{AC})$$

Function key: 0009

Purpose: To compute coordinates
of traverse stations

TRAVERS

Function: $(X_i, Y_i) = f(\alpha_{i-1}, \bar{p}_0, X_0, Y_0, S_i)$; see figure 4

Procedure:

1. Key Y₀ in Y; X₀ in X
2. Depress C9
3. Key $\bar{p}_0^*$ in X
4. Depress G0
5. Key S_i in Y; α_{i-1}^* in X
6. Depress G0
7. Read Y_i in Y; X_i in X

Repeat steps 5-7 for other values of S and α .

Example: See figure 4

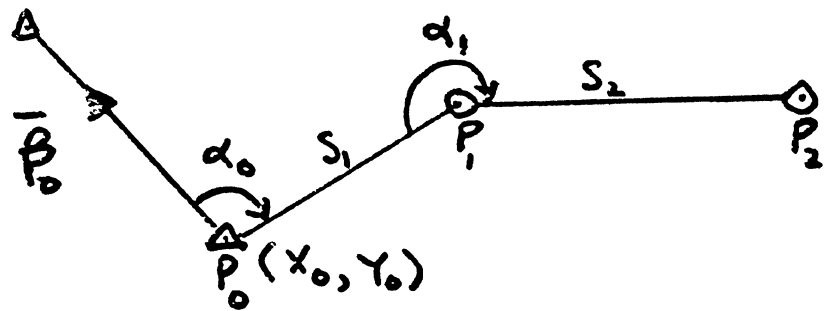
1. Key 10.0 in Y; 10.0 in X
2. Depress O9
3. Key 27.000000 in X
(270° 00' 00")
4. Depress G0
5. Key 500.000 in Y; 1200000 in X
(120° 00' 00")
6. Depress G0
7. Read 259.999999991 in Y; 443.012701911 in X

Repeat steps 5-7 to obtain coordinates of next point (that is point C in the figure)

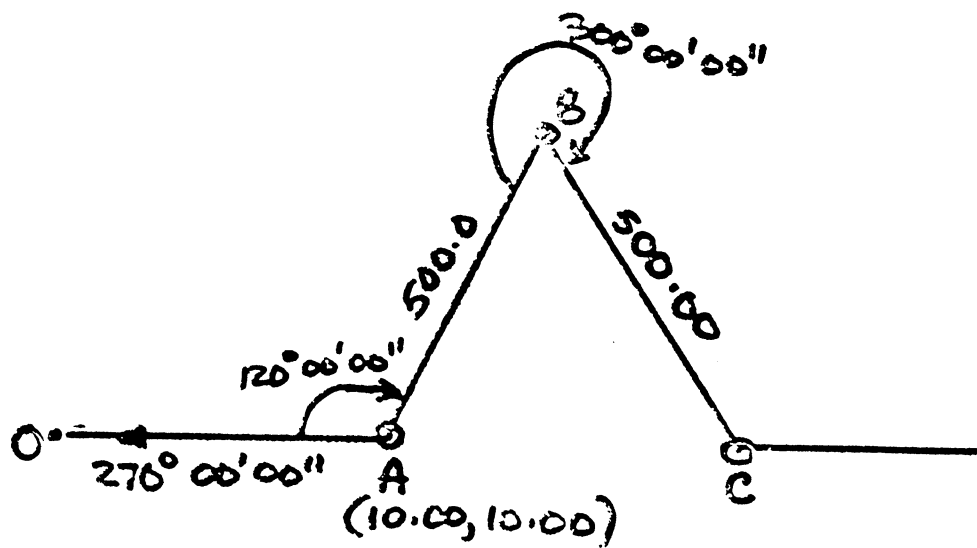
5. Key 210.00 in Y; 1000000 in X
(300° 00' 00")
6. Depress G0
7. Read 109.999999979 in Y; 10.0000000000 in X

* Values should always be in DMS(degrees, minutes and seconds) for this routine (TRAVERS).

FIGURE 4



EXAMPLE



Function Key: 0011

Purpose: To calculate spherical coordinates
(R, β , α) from cartesian coordinates
X, Y, Z. β , α in radians.

SPHER

(inverse to CART)

Function: 1. $(\alpha, \sqrt{X^2 + Y^2}) := \text{SPHER}(X, Y)$
2. $(\beta, R) := \text{GO}(Z, \sqrt{X^2 + Y^2})$

Procedure: 1. Key Y in Y and X in X register
2. Depress 0011
3. Read α (in radians) from X register, eventually convert
to degrees, minutes, seconds or grads.
4. Key Z in X register
5. Depress GO
6. Read R from Y and β (in radians) from X register

Example: 1. Key 767.576 in Y and -582.645 in X register.
2. Depress 0011
3. Depress DEG (0015) and read $127^\circ 12' 13''.93921$ from
X register
4. Key 267.117 in X register
5. Depress GO
6. Depress DEG (0015) and read $15^\circ 29' 34''.031275$ from
X and 999.999801744 from Y register.

Remark: R = slope distance
 β = vertical angle
 α = horizontal angle

Function key: 0012

POLAR

Purpose: To calculate polar
coordinates (R, θ)
from rectangular
coordinates (X, Y). θ
in radians.

Function: (R, θ) = POLAR (X, Y)

Procedure: Key Y coordinate in X; Depress $\uparrow$
Key X coordinate in X; Depress C12

Read: Distance R in Y

Angle θ in X (in radians)

Depress C15; Read θ in DMS in X

Example: Key 5 in X; Depress $\uparrow$

Key 4 in X; Depress C12

Read: 5.0000000000 in Y

.64351108792 radians in X

Depress C15; Read 365211.631524 in X
(36° 52' 11.631524")

Function key: 0013

ARCTAN

Purpose: To compute the arctan
of a value.

Function: X = ARCTAN (X)

Procedure: Key argument in X; Depress C13

Read arctan given in radians in X

Example: Key 1.0075821 in X; Depress C13

Read .7850075623072

Note: If result is wanted in degrees, minutes and seconds,
depress 1102.

Function key: 0014

GON

Purpose: To convert an angle
 given in radians to
 gons.

Function: $X = \text{GON } (X)$

Procedure: Key radians in X; Depress C14

Example: Key 3.14159265359; Depress C14
 Read 200.00000000 in X

Function key: 0015

DMS

Purpose: To convert an angle
 given in radians into
 a DMS-value (degrees,
 minutes, seconds)

Function: $X = \text{DMS } (X)$

Procedure: Key radians in X; Depress C15
 Read DMS-value in X

Example: Key 0.3781259; Depress C15
 Read 215954.065500
 (21° 39' 54" 065500)

000	MAIN		
1	RADEGR	0000	BEGIN RADEGR
2	↑		
3	WRITEA		
4	STDIR		
5	INTX		
6	STDIR		
7	120		
8	6		
9	0		
10	XDIR		
11	120		
12	↓		
13	WRITEA		
14	XDIR		
15	INTX		
16	+DIR		
17	120		
18	4		
19	0		
20	XDIR		
21	120		
22	REDIR		
23	120		
24	-		
25	π		
26	X		
27	6		
28	4		
29	8		
30	SEXP		
31	6		
32	÷		
33	↓		
34	SEARCH		END RADEGR
35	1515		
36	MAF		
37	RADGON	0001	BEGIN RADGON
38	↑		
39	π		
40	X		
41	2		
42	0		
43	0		
44	÷		
45	↓		
46	SEARCH		END RADGON
47	1515		
000	MAIN		
01	COS	0003	BEGIN COS
02	↑		
03	+		
04	π		
05	+		
06	2		
07	÷		
08	↓		

089	MARI		
90	SIN	442	BEGIN SIN
91	STDIR		
92	120		
93	π		
94	$\uparrow$		
95	+		
96	REDIR		
97	120		
98	EXY ($\uparrow\downarrow$)		
99	$\div$		
100	EXY ($\uparrow\downarrow$)		
1	INTX		
2	X		
3	REDIR		
4	120		
5	LXY ($\uparrow\downarrow$)		
6	-		
7	π		
8	$\div$		
9	$\downarrow$		
10	INTX		
11	WRITEA		
12	LOG _e X		
13	-		
14	CHSGN		
15	1		
16	X		
17	.		
18	5		
19	SI IPY < X		
20	SEARCH		
21	1508		
22	CHSGN		
23	SI IPY $\geq$ X		
24	SEARCH		
25	1508		
26	MARI		
27	1507		
28	π		
29	X		
30	STDY	414	
31	120		
32	$\downarrow$		
33	X ²		
34	STDIR		
35	113		
36	1		
37	5		
38	$\uparrow$		
39	1		
40	STDIR		
41	112		
42	MARI		
43	1506		
44	REDIR		
45	113		

146	CHSGN	
47	XDIR	
48	112	
49	↓	
50	÷DIR	
51	112	
52	1	
53	-	
54	↓	
55	÷DIR	
56	112	
57	1	
58	+DIR	
59	112	
60	RESID	
61	+DIR	
62	112	
63	1	
64	-	
65	STIPY-X	
66	SEARCH	
67	1506	
68	REDIR	
69	112	
70	RLDY	C415
71	120	
72	X	
73	↓	
74	SEARCH	
75	1515	
76	MARI	
77	1508	
78	-	
79	-	
80	↓	
81	CHSGN	
82	↑	
83	SEARCH	
84	1507	END COS AND SIN
85	MARI	
86	TAN	0004
87	STDY	C414
88	111	
89	↑	
90	0002	
91	EXY	
92	0003	
93	÷	
94	↓	
95	RETX	C415
96	111	
97	RETURN	
98	MARI	
99	RECT	C415
200	STDIR	
01	111	
02	0003	
03	EXY (↑↓)	

204	X		
05	EXDIR(↻)		
06	111		
07	0002		
08	XDIR		
09	111		
10	↓		
11	REDY	C415	
12	111		
13	RETURN		
14	MAR		
15	ARCTAN	0013	BEGIN ARCTAN
16	WRITEA		
17	CLX		
18	↑		
19	•		
20	5		
21	SIMPY>X		
22	WRITEA		
23	X ²		
24	1		
25	+		
26	STDY	C414	
27	113		
28	2		
29	-		
30	REDIR		
31	113		
32	÷		
33	↓		
34	STDY	C414	
35	112		
36	X		
37	STDY	C414	
38	113		
39	1		
40	STDIR		
41	120		
42	1		
43	5		
44	↑		
45	8		
46	STDIR		
47	111		
48	MAR		
49	1513		
50	REDIR		
51	113		
52	XDIR		
53	120		
54	REDIR		
55	111		
56	XDIR		
57	111		
58	EXDIR		
59	111		
60	REAR		

261	120		
62	↓		
63	+DIR		
64	120		
65	2		
66	-		
67	1		
68	-DIR		
69	111		
70	EXDIR		
71	120		
72	*DIR		
73	120		
74	REDIR		
75	111		
76	WRITEA		
77	LOG _e X		
78	SEARCH		
79	1513		
80	REDY	0415	
81	112		
82	REDX		
83	120		
84	X		
85	WRITEA		
86	GO		
87	X		
88	4		
89	5		
90	WRITEA		
91	10 ^x		
92	WRITEA		
93	STOP		
94	X		
95	↓		
96	WRITEA		
97	INDPRG		
98	SEARCH		
99	1515		END ARCTAN
300	MARK		
1	POLAR	012	BEGIN POLAR
2	WRITEA		
3	WRITE		
4	SEARCH		
5	1505		
6	WRITEA		
7	SETEXP		
8	IXY(↑↓)		
9	RETURN		
10	CHSGN		
11	↑		
12	TT		
13	RETURN		
14	MARK		
15	1505		
16	WRITEA		
17	LOG _e X		

318	SEARCH	
19	15C4	
20	1	
21	SLEXP	
22	CHSGN	
23	9	
24	5	
25	MARF	
26	15C4	
27	WRITLA	
28	e^x	
29	$\div$	
30	x^2	
31	STDIR	
32	11C	
33	REDIR	
34	121	
35	x^2	
36	+ DIR	
37	11C	
38	RIDIR	
39	11C	
40	EXY($\nwarrow$)	
41	WRITEA	
42	LOG10X	
43	1/X	
44	CHSGN	
45	C13	
46	STDIR	
47	11C	
48	π	
49	$\uparrow$	
50	2	
51	$\div$	
52	1	
53	WRITLA	
54	LOADPRG	
55	X	
56	REDIR	
57	11C	
58	+	
59	REDIR	
60	121	
61	$\sqrt{x}$	
62	EXY($\nwarrow$)	
63	RETURN	
64	MARF	
65	GON	C14
66	$\uparrow$	
67	2	
68	0	
69	0	
70	X	
71	π	
72	$\div$	
73	$\downarrow$	
74	SEARCH	
75	1515	

END POLAR

BEGIN GON

376	MARI		
77	DEGR	0015	BEGIN DEGR
78	↑		
79	WRITEA		
80	GO		
81	X		
82	CLX		
83	STDIR		
84	120		
85	0315		
86	0315		
87	RIDIR		
88	120		
89	+		
90	↓		
91	SEARCH		
92	1515		
93	MARI		
94	0315		
95	↓		
96	INTX		
97	-		
98	+ DIR		
99	120		
400	2		
01	10X		
02	XDIR		
03	120		
04	6		
05	0		
06	X		
07	RETURN		END DEGR
08	MARI		
09	ECCORR	0007	BEGIN ECCORR
410	STDY(0414)		
11	065		
12	STDIR		
13	064		
14	MARI		
15	GO		
16	STOP		
17	MARI		
18	1		
19	0314		
20	0005		
21	-DIR		
22	063		
23	RIDIR		
24	063		
25	0313		
26	MARI		
27	2		
28	0314		
29	0002		
30	SEARCH		
31	1500		
32	-447=064; 065		
440	-463=062; 063		

454	MAIN		
455	1500		
456	X		
457	REDIR		
458	063		
459	$\div$		
460	STDY(C414)		
461	063		
462	1		
463	EXY($\uparrow\downarrow$)		
464	X ²		
465	$\downarrow$		
466	$\sqrt{X}$		
467	RNDY(C415)		
468	063		
469	0313		
470	MARK		
471	0314		
472	STDY(C414)		
473	063		
474	STDIR		
475	062		
476	$\uparrow$		
477	REDIR		
478	064		
479	-		
480	$\downarrow$		
481	RNDY(C415)		
482	065		
483	RETURN		
484	MAIN		
485	0313		
486	012		
487	+ DIR		
488	062		
489	π		
490	$\uparrow$		
491	2		
492	X		
493	$\downarrow$		
494	RNDY(C415)		
495	062		
496	SKIPY < X		
497	-		
498	GO		
499	CLX		
500	EXY($\downarrow\uparrow$)		
501	SEARCH		
502	GO		
503	MAIN		
504	1515		
505	RNDY(C415)		
506	121		
507	RETURN		END ECCORR
508	MARK		
509	CART	006	BEGIN CART

521	0005		
522	EXY		
523	STOP		
524	0005		
525	STOP		END CART
526	MARK		
527	SPHER	0011	BEGIN SPHER
528	012		
529	STOP		
530	EXY		
531	012		
532	STOP		END SPHER
533	MARK		
534	INSECT	0008	BEGIN INSECT
535	1		
536	↑		
537	STOP		
538	STDY		
539	062		
540	STDY	0414	
541	063		
542	2		
543	↑		
544	STOP		
545	EXDX		
546	062		
547	-DX		
548	062		
549	EXY		
550	EXDX		
551	063		
552	-DX		
553	063		
554	EXDX		
555	063		
556	EXY		
557	EXDX		
558	062		
559	POLAR	0012	
560	STDY		
561	064		
562	STDY	0414	
563	065		
564	1		
565	↑		
566	2		
567	x		
568	STOP		
569	-DX		
570	064		
571	EXY		
572	-		
573	EXDX		
574	064		
575	0002		
576	EXY		

577	CHSGN		
578	0002		
579	✓		
580	REDX		
581	065		
582	*		
583	REDX		
584	064		
585	0005		
586	+DX		
587	062		
588	REDX		
589	063		
590	+		
591	REDX		
592	062		
593	STOP		END INSECT
594	MARK		
595	TRAVERS	0009	BEGIN TRAVERS
596	STDY		
597	001		
598	STDY	0414	
599	002		
600	STOP		
601	RADEG	0000	
602	STDY		
603	000		
604	PI		
605	+DX		
606	000		
607	MARK		
608	STOP		
609	STOP		
610	RADEG	0000	
611	+DX		
612	000		
613	PI		
614	-DX		
615	000		
616	REDX		
617	000		
618	RECT	0005	
619	+DX		
620	001		
621	↓		
622	+DX		
623	002		
624	REDY	0415	
625	002		
626	REDX		
627	001		
628	SEARCH		
629	STOP		
630	ENDPRGM		END TRAVERS; END SURVTRIGPAK